

# Paperfly Courier API — Integration Guide

Source: Paperfly Merchant Panel → Developer Guide

(<https://go.paperfly.com.bd/merchant/developer-guide>), captured 2026-09-09.

This document is meant to be handed to a coding assistant (e.g. Claude Code) as the spec for building a Paperfly courier integration into a backend service (NestJS/Node.js + TypeScript recommended).

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## 1. Overview

Paperfly exposes a small REST API for:

- Creating a delivery order (parcel)
- Creating an exchange order
- Tracking an order
- Cancelling an order
- Webhooks for real-time status updates (parcel lifecycle events)
- A WooCommerce plugin (not relevant for a custom backend integration)

All order-management endpoints use **HTTP Basic Auth** with the merchant's **Paperfly Merchant Panel username and password**, plus a fixed API key sent as a custom header.

 **Credentials note:** the Basic Auth username/password are the actual Paperfly merchant panel login credentials — treat them as secrets (env vars / secret manager), never hard-code or commit them. The same goes for the **paperflykey** header value below.

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## 2. Authentication

- **Type:** HTTP Basic Auth
  - Username: Merchant Panel username
  - Password: Merchant Panel password
- **Required header on Order Creation / Exchange Order:**
  - **paperflykey:** Paperfly\_~La?Rj73FcLm
  - **Content-Type:** application/json

(Tracking and Cancellation endpoints only show Basic Auth in the docs — no **paperflykey** header listed for those, but it's worth sending it anyway in case it's required server-side and just undocumented.)

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## 3. Endpoints

### 3.1 Create Order (single parcel)

POST [https://api.paperfly.com.bd/merchant/api/service/new\\_order\\_v2.php](https://api.paperfly.com.bd/merchant/api/service/new_order_v2.php)

#### Headers

Content-Type: application/json

paperflykey: Paperfly\_~La?Rj73FcLm

Authorization: Basic <base64(username:password)>

#### Request body

```
{  
  
  "merchantOrderReference": "Test_01610",  
  
  "storeName": "Ovi",  
  
  "productBrief": "Test Product",  
  
  "packagePrice": "10",  
  
  "max_weight": "0.3",  
  
  "customerName": "Liton Ovi",  
  
  "customerAddress": "Banani, Dhaka",  
  
  "customerPhone": "01610202717"  
}
```

## Success response

```
{  
  
  "success": {  
  
    "message": "successfully inserted",  
  
    "tracking_number": "Z-051125-63821-A3-A1",  
  
    "tracking_barcode": "751820115459"  
  
  },  
  
  "response_code": 200  
}
```

| Field                  | Type   | Notes                                                   |
|------------------------|--------|---------------------------------------------------------|
| merchantOrderReference | string | Your own unique order ID (idempotency key on your side) |
| storeName              | string | Store/shop name registered with Paperfly                |
| productBrief           | string | Short product description                               |
| packagePrice           | string | COD amount / package value                              |
| max_weight             | string | Weight in kg                                            |
| customerName           | string | Recipient name                                          |
| customerAddress        | string | Recipient address                                       |
| customerPhone          | string | Recipient phone (BD format, e.g. 01XXXXXXXXXX)          |

## 3.2 Create Exchange Order

Same endpoint as order creation, with extra exchange-specific fields.

POST [https://api.paperfly.com.bd/merchant/api/service/new\\_order\\_v2.php](https://api.paperfly.com.bd/merchant/api/service/new_order_v2.php)

## Request body

```
{  
  
  "merchantOrderReference": "Test_01615",  
  
  "storeName": "Ovi",  
  
  "productBrief": "Test Product",  
  
  "packagePrice": "10",  
  
  "max_weight": "0.3",  
  
  "customerName": "Liton Ovi",  
  
  "customerAddress": "Banani, Dhaka",  
  
  "customerPhone": "01610202717",  
  
  "orderType": "Exchange",  
  
  "exchangeDescription": "exchange product",  
  
  "exchangePrice": "100",  
  
  "exchangeWeight": "1.5"  
}
```

Additional fields when **orderType** is "Exchange":

| Field                      | Meaning                              |
|----------------------------|--------------------------------------|
| <b>orderType</b>           | "Exchange"                           |
| <b>exchangeDescription</b> | Description of the exchange product  |
| <b>exchangePrice</b>       | Exchange amount                      |
| <b>exchangeWeight</b>      | Package weight for the exchange item |

### 3.3 Track Order

POST <https://api.paperfly.com.bd/API-Order-Tracking>

**Headers:** Basic Auth (Merchant Panel username/password)

#### **Request body**

```
{ "ReferenceNumber": "Test_01610" }
```

(Use the [merchantOrderReference](#) you submitted at order creation.)

#### **Success response**

```
{  
  
  "success": {  
  
    "message": "success",  
  
    "trackingStatus": [  
  
      {  
  
        "invNum": "",  
  
        "receivedAmount": "",  
  
        "Pick": null,  
  
        "PickTime": null,  
  
        "inTransit": "",  
  
        "inTransitTime": "",  
  
        "ReceivedAtPoint": "",  
  
        "ReceivedAtPointTime": "",  
  
        "PickedForDelivery": "",  
  

```

```

        "PickedForDeliveryTime": "",
        "Delivered": "",
        "DeliveredTime": "",
        "Returned": "",
        "ReturnedTime": "",
        "Partial": "",
        "PartialTime": "",
        "onHoldSchedule": "",
        "close": "",
        "closeTime": ""
    }
]
},
"response_code": 200
}

```

Each field is populated with a status/timestamp once that lifecycle stage happens (empty string / **null** until then).

### 3.4 Cancel Order

POST <https://api.paperfly.com.bd/api/v1/cancel-order>

**Headers:** Basic Auth (Merchant Panel username/password)

#### **Request body**

```
{ "order_id": "Test_01610" }
```

## Success response

```
{  
  
  "success": {  
  
    "message": "succesfully canceled",  
  
    "response_code": 200  
  
  }  
  
}
```

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## 4. Webhooks (real-time status push)

Configured from the merchant panel (Developer Guide → Webhooks → Configuration): you register a **Webhook Endpoint URL** and a **Secret Key**, then select which events to subscribe to.

- The secret key is sent in the request header of each webhook call, for signature/verification on your side (exact header name wasn't shown in the UI text — check the panel or ask Paperfly support for the header name).
- Paperfly sends a **POST** request to your endpoint with a JSON payload when a subscribed event fires.
- Your endpoint should return **HTTP 200** to acknowledge receipt.
- Timeout: **30 seconds**. Failed deliveries are **retried up to 3 times**.
- All timestamps are ISO 8601 (UTC).

## Available events

### Order Lifecycle

- **parcel.created** — parcel initially placed
- **parcel.invoiced** — invoice generated
- **parcel.cancelled** — order deleted/cancelled

### Pickup & Transit

- **parcel.picked\_up** — parcel picked up (handshake)
- **parcel.in\_transit** — parcel in transit

- `parcel.received_at_point` — received at local point (incl. Pathao point)

## Delivery

- `parcel.assigned_for_delivery` — out for delivery
- `parcel.delivered` — successfully delivered
- `parcel.partial` — partially delivered
- `parcel.exchange` — exchange order completed
- `parcel.on_hold` — delivery on hold

## Returns

- `parcel.return` — returned delivery action
- `parcel.return_transit` — in transit for return (reverse journey)
- `parcel.return_to_merchant` — returned to merchant

## Payload example

```
{  
  
  "event": "parcel.created",  
  
  "timestamp": "2025-12-24T17:28:24+00:00",  
  
  "data": {  
  
    "order_number": "Z-241225-174131-A1-A7",  
  
    "merchant_order_reference": "test12121212",  
  
    "barcode": "231814375965",  
  
    "package_price": 10,  
  
    "recipient": {  
  
      "name": "Shamim Ahammed Shamim",  
  
      "phone": "01685048848",  
  
      "address": "house 1/2, road 6, Kaderabad Housing, Mohammadpur"  
    },  
  },  
}
```

```
"special_instruction": ""  
  
}  
  
}
```

`data` shape likely varies per event type (e.g. delivery/return events probably carry different fields) — validate loosely (e.g. with a Zod schema that only requires `event`, `timestamp`, and a generic `data` object) rather than assuming this exact shape for every event.

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## 5. Suggested NestJS integration shape

For a NestJS + TypeScript backend, a clean integration typically looks like:

- `PaperflyModule`
    - `PaperflyService` — thin HTTP client (axios/`HttpService`) wrapping the 4 endpoints above, reading base URL / credentials / `paperflykey` from `ConfigService` (env vars: `PAPERFLY_USERNAME`, `PAPERFLY_PASSWORD`, `PAPERFLY_API_KEY`).
    - `PaperflyController` (optional) — internal endpoints your own frontend calls to create/track/cancel a shipment, which delegate to `PaperflyService`.
    - `PaperflyWebhookController` — a `POST /webhooks/paperfly` route that:
      - verifies the secret key from the configured header,
      - validates payload with a DTO/Zod schema,
      - maps `event` to your own order status enum,
      - is idempotent (dedupe by `merchant_order_reference` + `event` + `timestamp`, since Paperfly retries failed deliveries up to 3×),
      - responds `200` quickly and does heavier processing (DB update, notifications) via a queue (BullMQ, since that's already in your stack) rather than inline.
    - DTOs for `CreateOrderDto`, `CreateExchangeOrderDto`, `TrackOrderDto`, `CancelOrderDto` mirroring the field tables above, validated with `class-validator`.
    - Retry/backoff + timeout handling on outbound calls (axios interceptor or `HttpService` config), since Paperfly's own webhook timeout is 30s — your outbound calls to Paperfly should also fail fast and retry sensibly.
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## 6. Open items to confirm with Paperfly / in the panel before go-live

- Exact header name Paperfly uses to send the webhook secret key (for signature verification).
- Whether **paperflykey** is also required on Tracking/Cancellation calls (docs only show it for order creation).
- Rate limits, if any.
- Sandbox/test credentials vs production credentials (the merchant panel shown belongs to account **sellerone | C-1-80610**).
- Bulk order creation endpoint (the panel has a "Create Multiple Parcel" UI action, but no separate bulk API was documented on this page — worth asking support, or check the WooCommerce plugin's source under "View Files & Documents" for hints).